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Asymmetric Model of Galactic Fountains for the Origins of Milky Way's Intermediate– and High–Velocity Clouds

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Abstract and Preliminary Results

The Milky Way's disk is surrounded by a layer of extraplanar gas, seen in all-sky neutral hydrogen (HI) surveys as diffuse emission at the disk–halo interface and as intermediate- and high-velocity clouds (IVCs and HVCs). While HVCs are mostly considered to be clouds accreting onto the Galaxy by the community, at least the IVC part of this gas is thought to originate in a galactic fountain. This is material lifted from the disk by stellar feedback, which travels through the halo and falls back. In the fountain-accretion model, the clouds also interact with the hot circumgalactic medium (CGM), part of which condenses onto them and is carried back. Still, this is a proposed model, not a general truth, and whether a fountain can produce HVCs at all remains controversial.

For over 50 years, why these clouds are so asymmetric between the northern and southern hemispheres has been a mystery, and mirror-symmetric fountain models cannot address it. Recently, eROSITA X-ray observations have shown that the hot CGM is itself asymmetric between the hemispheres, including in its temperature, and simulations of the Milky Way–Magellanic Clouds interaction (e.g., Oprea et al. 2026) explain this as a motion of the Milky Way relative to its hot halo. Such a motion changes the pressure of the hot gas (hence the drag it exerts) and its speed relative to the clouds in each hemisphere, so it could also shape the cold extraplanar gas.

The goal of this thesis is to explain the IVC/HVC asymmetry with a dynamical galactic-fountain model of the Milky Way that includes these effects, fitted to all-sky LAB HI data (HI4PI planned). Secondarily, it lets us test whether the fountain can produce HVCs and whether it also promotes accretion from the hot CGM, and at what rate. To my knowledge, this would be the first application of a non-symmetric fountain model to the Milky Way.

So far, I have adapted the existing fountain code from Urbina et al. (in prep.), based on Fraternali & Binney (2008) and Li et al. (2023), to the Milky Way. I implemented the projection that produces all-sky datacubes as seen from the Sun and validated them by reproducing the benchmark results of Marasco et al. (2012). To assess how robustly parameters can be recovered, I characterized the LAB noise and tested model numerical convergence. I ran a grid search (Universal Kriging Gaussian Process) fitting the model to LAB, with one parameter set shared by both hemispheres as a symmetric baseline, and examined how different definitions of the model–data mismatch affect the conclusions. It reveals trends in parameter space, and the expected preference for condensation-driven accretion is recovered robustly. You can see my results so far in the repository.

The next stage is to implement separate treatment of the hemispheres, so the hot CGM can exert different drag and have a different relative speed in each. Over the same parameter space, I will then run a full Bayesian inference (MCMC) and use Bayes factors and the Bayesian Information Criterion to assess whether, and by how much, breaking the symmetry improves the description of the data. Beyond this question, the framework offers a route to connect the HI and X-ray views of the Galaxy's halo and to quantify how the interaction with the Magellanic Clouds has shaped the Milky Way's current gas cycle.